

Elliptic Curve Cryptography

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Academic Year: 2020-21

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1 Cryptography

1.1 Basic Terminology

Definition 1.1.1. *The raw message that is sent from a sender to a receiver is called Plaintext.*

Definition 1.1.2. *Encryption is the process of disguising the plaintext in order to hide it's content from an intruder.*

Definition 1.1.3. *The new message obtained after encrypting a plaintext is called Ciphertext.*

Definition 1.1.4. *Decryption is the process of converting the Ciphertext back into Plaintext.*

Definition 1.1.5. *The science or study of keeping messages secure and protected is called Cryptography. Cryptography is practiced by Cryptographers¹.*

If a message or plaintext is denoted by 'M', the ciphertext by 'C' and the functions of encryption and decryption by 'E' and 'D' respectively, then these processes can be mathematically written as:

$$\begin{aligned} E(M) &= C \\ D(C) &= M \end{aligned}$$

or, the entire process of sending a secure message can be written in it's most simplest form as:

$$D(E(M)) = M$$

or,

$$E(M) = D^{-1}(M)$$

In addition to the confidentiality of a secret message sent, cryptography is expected to adhere to the following rules as well.

¹All definitions in this section (1.1) is referred from [2]

Authenticity - The receiver must be certain of the origin from where the message is being sent and that no intruder is involved.

Integrity - The receiver must be sure that the message sent by the sender has not been modified by an intruder during its transit.

Non-repudiation - The sender cannot deny later that the message was sent by him.²

1.2 History

The history of cryptography goes back a long way during 1900 BC when the first ciphertext was discovered in the chamber of the nobleman Khnumhotep II, in Egypt [13]. From then onwards there has been many improvements and advancements in cryptography. Some of the historical ones that are most known are the *Caesar Cipher* and the *Enigma Machine*. [1]

1.2.1 Caesar Cipher

This cipher was used by and named after *Julius Caesar* (12 July 100 BC – 15 March 44 BC), a Roman General and statesman, to convey secret messages to his army generals and for other military significance. Caesar cipher is a *substitution cipher*. A substitution cipher is when every character in a plaintext is substituted by another character in ciphertext to encrypt the message. The receiver of the message then decrypts it by just inverting the substitutions.

In Caesar cipher, each plaintext character is substituted by the character 3 to right modulo 26. This means that ‘A’ is written as ‘D’, ‘B’ is written as ‘E’, ... , ‘X’ is written as ‘A’, ‘Y’ is written as ‘B’ and ‘Z’ is written as ‘C’. This is a one of the most simplest forms of ciphers.

1.2.2 Enigma Machine

During the beginning of the 19th century, when everything became electric, Edward Hebern invented the electro-mechanical machine called Hebern rotor machine which consisted a rotor in it to automate the process of encryption and decryption. The rotor machine has a keyboard and rotors that have an arbitrary permutation of the alphabets, 26 positions and performs substitution on each character. Many rotor machines have multiple rotors with

²[2, Chp1, P2]



Figure 1: Enigma Machine

different permutations to substitute different characters. Here, the output pin of one rotor is connected with the input pin of the next rotor.

One of the best known rotor devices used is the Enigma machine. The Enigma was used by Germans during World War II to send secret messages to the military troops. The Enigma had three rotors that were randomly chosen from a group of five, a reflecting rotor which caused each rotor to perform a substitution on a plaintext twice and a plugboard for further permuting the plaintext. This made the Enigma very complicated and secure. Although, it was broken by a team of Polish cryptographers who informed the British intelligence. Then the famous Mathematician and Cryptographer, Alan Turing, built the *Bombe machine* which cracked the key to the Enigma machine which was set to change daily.

2 Public Key Cryptography

2.1 Cryptographic Keys

Definition 2.1.1. *The mathematical function which is used for encryption and decryption is called a cryptographic algorithm or a cipher.*

When the working of an algorithm is kept as a secret for its security, the algorithm is then a *restricted algorithm*.

Definition 2.1.2. *A set of data or piece of information that is used by the cryptographic algorithm to transform the plaintext to ciphertext or vice versa*

is called a *key*.³.

The pair of keys mainly used in the Public Key Encryption (2.2) explained below are the private and public keys. **Private keys** are keys which is only known and used by the owner. It is kept confidential from the public. **Public keys**, as the name suggests, is known to the public and is openly accessible.

2.2 Public Key Encryption

There are two key-based algorithms. One is the symmetric algorithm and the other is the public-key or asymmetric algorithm.

Symmetric Encryption

In symmetric encryption, one common key is used to encrypt and decrypt the secret message. The sender and receiver must secretly agree on the common key for their secure communication. The security of this algorithm completely depends on the shared key. If any intruder gets hold of the key, the entire purpose of the algorithm would be lost. If 'k' is denoted as the common key,

$$\begin{aligned}E_k(M) &= C \\D_k(C) &= M\end{aligned}$$

Public Key Encryption

In contrast to symmetric encryption, the key that is used for encryption and decryption are different. Here, the sender and receiver own a pair of keys, the public key and private key. The encryption key is usually called the public key while the decryption key is called the private key. The private keys are kept as secret while the public keys are openly exchanged between the sender and receiver to confirm authenticity of the message. Let ' k_1 ' and ' k_2 ' be two different keys. Then,

$$\begin{aligned}E_{k_1}(M) &= C \\D_{k_2}(C) &= M\end{aligned}$$

Let's say Alice wants to securely send a message to Bob. If Alice uses *symmetric encryption*, she and Bob must first securely share the common key

³The definitions in this section (2) is referred from [2]

that they would use. Then Alice would first encrypt the message and send it over to Bob who would then decrypt it using the same key and read message.

Now, if instead they decide to use *public key encryption*. Alice and Bob both have one private and one public key each. First, Alice can encrypt her message using Bob's public key, and send it over to him. Bob can decrypt this using his own private key and read the message. This makes sure that the message is sent confidentially without having to exchange the key previously.

Next, Alice can encrypt the message with her own private key and send it to Bob, who can decrypt it using Alice's public key. Anyone can decrypt and access this message, but, it is certain that the message was sent authentically by Alice and no one else.

Hence, to make sure the message is confidential and authentic, Alice first encrypts it with her own private key, then again encrypts it using Bob's public key. Once the message reaches Bob, he first decrypts it using his own private key and then using Alice's public key. This algorithm based on a pair of keys is called *Public Key Encryption*.

Examples of Public Key Encryption include Diffie Hellman Key Exchange, RSA Cryptography, ElGamal Encryption, Elliptic Curve Cryptography, etc. In this essay, I will dive into the details of Elliptic Curve Cryptography.

2.3 Diffie Hellman Key Exchange

Before I begin discussing about Elliptic Curve Cryptography, I have lightly touched upon the concept of Diffie Hellman key exchange. This public key encryption is one of the many public key encryption protocols. In 1976, Whitfield Diffie and Martin Hellman published their work, marking the beginning of public key encryption in public [8]. In their published paper, they proposed the multiplicative group of a finite prime field.

First, we will describe this exchange using a cyclic finite group, G of order n . Let g be the generator in group G . The values of g and n are publicly accessible values and hence known by any attacker. Now, let's say that Alice wants to send a message to Bob. So, Alice chooses a random natural number

a , such that, $1 < a < n$. This would be her private key. Similarly, Bob chooses a random natural number b , such that $1 < b < n$ which would be his private key. Alice and Bob calculate g^a and g^b , respectively, which are their public keys.

As discussed before in section 2.2, for Alice to send a message to Bob, she encrypts using her own private key and Bob's public key. So, Alice computes $(g^b)^a = g^{ab}$. Similarly, Bob should decrypt the message using his own private key and Alice's public key. Hence, Bob computes $(g^a)^b = g^{ab}$. Thus, g^{ab} is now the shared secret key between Alice and Bob that they would use to securely encrypt and decrypt messages sent between them.

We can also provide the original cryptographic explanation of this key exchange using an example. This is explained by multiplicative group of integers modulo n , where, n is prime. Let g be the primitive root modulo n . Thus, the shared secret key would be between 1 and $1 - n$. (Note: Any variables below in bold means that it is private or secret while the rest of the variables are public or non-secret).

Let,

$$g = 13$$

$$n = 55$$

$$\mathbf{a} = 9 \text{ (Alice's private key only known by Alice)}$$

$$\mathbf{b} = 10 \text{ (Bob's private key only known by Bob)}$$

$$A = g^{\mathbf{a}} \bmod n = 28 \text{ (Alice's public key)}$$

$$B = g^{\mathbf{b}} \bmod n = 34 \text{ (Bob's public key)}$$

$$\text{Alice then computes } B^{\mathbf{a}} \bmod n = 34 = \mathbf{s} \text{ (say)}$$

$$\text{Bob computes } A^{\mathbf{b}} \bmod n = 34 = \mathbf{s}$$

Hence, $\mathbf{s}$ is the secret shared key between Alice and Bob. The computations are both equal because,

$$A^{\mathbf{b}} \bmod n = g^{\mathbf{ab}} \bmod n = g^{\mathbf{ba}} \bmod n = B^{\mathbf{a}} \bmod n$$

The integer size used in this example is short (64-bits) compared to the 2048-bits or more originally used, making the keys long and complicated in contrast to the sizes in the currently used encryptions and Elliptic Curve

Cryptography. The attacker in the public does not have any knowledge of a and b , or the private keys, and thus finds it infeasible to calculate the secret key with only the given public information. Hence, making this a secure way of communicating. This protocol is the *Diffie Hellman Key Exchange*.

3 Elliptic Curves

Definition 3.1. *An elliptic curve E over a field K is defined by the following equation:*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6 \quad (1)$$

where parameters $a_1, a_2, a_3, a_4, a_6 \in K$. So, E is defined over its underlying field K . This equation is referred to as the *Weierstrass equation* for elliptic curves. Elliptic curves are two dimensional curves that are symmetric about the x-axis. The shape of the curve may vary depending upon the parameters of the curve.

The discriminant of equation (1) is defined as:

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6 \quad (2)$$

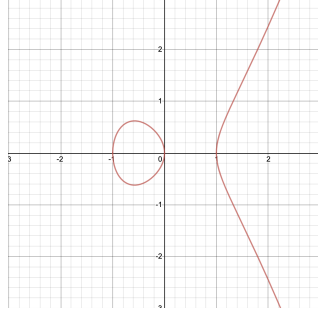
where,

$$\begin{aligned} b_2 &= a_1^2 + 4a_2 \\ b_4 &= 2a_4 + a_1a_3 \\ b_6 &= a_3^2 + 4a_6 \\ b_8 &= a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2 \end{aligned}$$

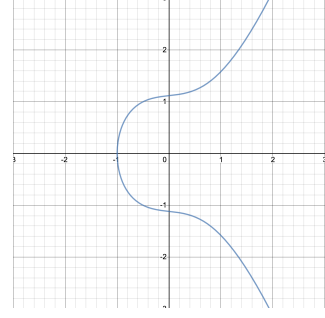
Now, an elliptic curve defined over K is a cubic equation, where $\Delta \neq 0$.
or,

$$-b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6 \neq 0 \quad (3)$$

This avoids singular points or any intersection in the curve. It also ensures ‘smoothness’, that is, there is no point on the curve that has two or more



(a) $y^2 = x^3 - x$



(b) $y^2 = x^3 + \frac{1}{4}x + \frac{5}{4}$

Figure 2: Elliptic curves over $\mathbb{R}^2$

distinct tangents.

We can simplify the Weierstrass equation (1) by making certain substitutions. First, let $\text{char}(K) \neq 2$. We replace y by $\frac{1}{2}(y - a_1x - a_3)$. This gives us,

$$y^2 = 4x^3 + b_2x^2 + 2b_4x + b_6 \quad (4)$$

Next, let $\text{char}(K) \neq 3$. We replace (x, y) by $(\frac{x-3b_2}{36}, \frac{y}{108})$. This results in,

$$y^2 = x^3 - 27c_4 - 54c_6, \quad (5)$$

where,

$$\begin{aligned} c_4 &= b_2^2 - 24b_4 \\ c_6 &= -b_2^3 + 36b_2b_4 - 216b_6 \end{aligned}$$

So, when $\text{char}(K)$ is not 2 or 3, we have,

$$1728\Delta = c_4^3 - c_6^2 \quad (6)$$

Considering the cubic $y^2 = x^3 + ax + b$ over K , if $\text{char}(K)$ is not 2 or 3, we have $c_4 = -48a$ and $c_6 = -864b$. So,

$$\Delta = \frac{(-48a)^3 - (-864b)^2}{1728} = -16(4a^3 + 27b^2) \quad (7)$$

This, by assuming that $\text{char}(K) \neq 2$ and $\text{char}(K) \neq 3$, the normal form of an elliptic curve over K is given by,

$$y^2 = x^3 + ax + b \quad (8)$$

We know that $\Delta \neq 0$ in elliptic curves. So, in it's simplified form we can say that,

$$\Delta = -16(4a^3 + 27b^2) \neq 0, \text{ or,} \\ (4a^3 + 27b^2) \neq 0$$

Any point (x, y) which satisfies equation (8) is a point on the curve E or $E(K) = \{(x, y) : y^2 = x^3 + ax + b\} \cup \{\infty\}$. (∞ is called the 'Point at infinity' [3, 5, 7]. Throughout this essay ' ∞ ' will refer to the 'Point at infinity'.)

3.1 The Group Law

Let E be an elliptic curve defined over the field K . The set of all the points in the elliptic curve that is defined over the underlying field K is $E(K) = \{(x, y) \in E : x, y \in K\} \cup \{\infty\}$. There is a *chord and tangent rule* that allows us to add two points in $E(K)$ to provide a third point in $E(K)$.

We start with two points, $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ on an elliptic curve E defined as $y^2 = x^3 + ax + b$. The addition rule of points on elliptic curves can be easily explained geometrically. Let's say that a line L passes through points P and Q on the curve E (Fig:3a). To obtain the sum of the two points, L is extended such that it intersects a third point on E and is then reflected across the x-axis. Let this reflected third point be called R . So,

$$P + Q = R \quad (9)$$

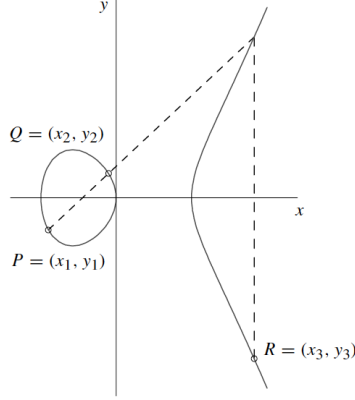
where, $R = (x_3, y_3)$

First, let's assume that $P \neq Q$ and neither of the points is the point at infinity. Then, the slope of line L would be,

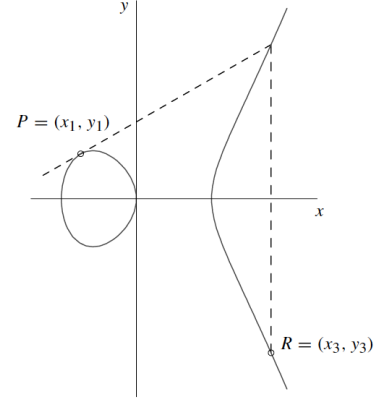
$$m = \frac{y_2 - y_1}{x_2 - x_1} \quad (10)$$

If $x_1 \neq x_2$, the equation of line L would be,

$$y = m(x - x_1) + y_1 \quad (11)$$



(a) Point Addition: $P + Q = R$



(b) Point Doubling: $P + P = R$

Figure 3: Geometric addition and doubling of points on Elliptic curve E

To find the intersection with E , we can substitute this in the elliptic curve equation,

$$(m(x - x_1) + y_1)^2 = x^3 + ax + b \quad (12)$$

This can be further rearranged as,

$$0 = x^3 - m^2x^2 + \dots \quad (13)$$

The three roots of the above cubic equation are the three points of intersection of line L and curve E . Let the roots of the cubic equation (13) be r, s, t . Calculating the roots here is not hard as two of the roots are already known, that is, points P and Q . We can find out the root as,

$$x^3 + ax^2 + bx + c = (x - r)(x - s)(x - t) = x^3 - (r + s + t)x^2 + \dots \quad (14)$$

Hence,

$$r + s + t = -a \quad (15)$$

If we already know r, s then we can calculate t by $t = -(a + r + s)$. So, we get,

$$x = m^2 - x_1 - x_2 \quad (16)$$

and

$$y = m(x - x_1) + y_1 \quad (17)$$

Now, the point $R = (x_3, y_3)$ would be the reflection of the above point. So,

$$\begin{aligned}x_3 &= m^2 - x_1 - x_2, \\y_3 &= m(x_1 - x_3) - y_1\end{aligned}\tag{18}$$

If $x_1 = x_2$, but, $y_1 \neq y_2$, the line L through P and Q would be vertical, intersecting curve E at ∞ . Reflecting this, the line would again intersect the curve at ∞ . So, in this case,

$$P + Q = \infty\tag{19}$$

Now, if we consider the case where $P = Q = (x_1, y_1)$, then the line L passing through P and Q would be a tangent of the curve E (Fig:3b). To find the slope of this tangent, we can use implicit differentiation.

$$\begin{aligned}2y \frac{dy}{dx} &= 3x^2 + a \\m &= \frac{dy}{dx} = \frac{3x_1^2 + a}{2y_1}\end{aligned}\tag{20}$$

If $y_1 = 0$, the line is vertical and we hence set, $P + Q = \infty$.
Now, if $y_1 \neq 0$, the equation of the tangent L is,

$$y = m(x - x_1) + y_1\tag{21}$$

This can be further rearranged as,

$$0 = x^3 - m^2x^2 + \dots\tag{22}$$

This time we only know one root. But, since the line is a tangent, it has a double root. We would, thus, only need to calculate one root. So, following the same steps as before, we would obtain,

$$\begin{aligned}x_3 &= m^2 - 2x_1, \\y_3 &= m(x_1 - x_3) - y_1\end{aligned}\tag{23}$$

If we consider a case where $Q = \infty$. Then the line L would be vertical passing through P and ∞ . The third point that this line intersects is the reflection

of point P over the x-axis. To find the sum of points P and ∞ , we need to again take the reflection of this third point which brings us back to P . Thus,

$$P + \infty = P \quad (24)$$

From the entire discussion above, we can form the following conclusions for the *Group Law*.

The Group Law

Let E be an elliptic curve defined by $y^2 = x^3 + ax + b$. Let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be points on E such that $P, Q \neq \infty$. $P + Q = R = (x_3, y_3)$ can be defined as follows:

1. If $x_1 \neq x_2$, then $x_3 = m^2 - x_1 - x_2$, $y_3 = m(x_1 - x_3) - y_1$ where, $m = \frac{y_2 - y_1}{x_2 - x_1}$. (Point Addition)
2. If $x_1 = x_2$ but $y_1 \neq y_2$, $P + Q = \infty$.
3. If $P = Q$ and $y_1 \neq 0$, then $x_3 = m^2 - 2x_1$, $y_3 = m(x_1 - x_3) - y_1$ where, $m = \frac{3x_1^2 + a}{2y_1}$. (Point Doubling)
4. If $P = Q$ and $y_1 = 0$, then $P + Q = \infty$
5. Lastly, $P + \infty = P$.

Theorem 3.1.1. *All the points on an elliptic curve E over the field K , or all the points in $E(K)$ form an additive abelian group.*

Proof. • Commutativity: This is obvious from the fact that any line that passes through P and Q is the same line that passes through Q and P . So, $P + Q = Q + P$ for all P, Q on E .

• Existence of identity: From the explanation preceding equation (24) and conclusion 5 under the group law, we can say if there is any point P on E , then $P + \infty = P$. As adding the point to ∞ gives the same point itself, ∞ is the identity element.

• Existence of inverse: If P' is the reflection of point P about the x-axis such that P, P' are on E , then the x-coordinates of the two points are equal. From conclusion 2 under the group law, we can say that $P + P' = \infty$. As ∞ is the identity element, P' is the inverse of P . This point P' is denoted as $-P$.

- **Associativity:** This is defined as $(P+Q)+R = P+(Q+R)$ for all P, Q, R on E . The associative property can be verified by a lengthy computation using explicit formulas or by using more advanced algebraic or analytic methods which is out of scope of this essay⁴.

Thus, all the points on an elliptic curve E form an *additive abelian group*. $\square$

If P is a point on an elliptic curve E and k is a positive integer, then, $P + P + P + \dots + P$, k times is denoted as kP . If $k < 0$, then the sum is written as, $kP = (-P) + (-P) + (-P) + \dots + (-P)$. If k is a very large number, we use *successive doubling*. This is also called as *scalar multiplication*. For example, to calculate $35P$, we first calculate $2P$, then $4P = 2P + 2P$, $8P = 4P + 4P$, $16P = 8P + 8P$, $32P = 16P + 16P$, $35P = 32P + 2P + P$.

Thus, we can write the process of successive doubling in general as, $kP = k_0P + k_1 \cdot 2P + k_2 \cdot 2^2P + \dots + k_r \cdot 2^rP$ with $k_0, k_1, \dots, k_r \in \{-1, 0, 1\}$. In this group, point P would be the *generator* of the group. Over the rational numbers, the numbers could grow rapidly, causing difficulty. However over a finite field, we work with the modulus, keeping the numbers involved relatively small.

3.2 Elliptic Curves over Finite Fields

Group Order

Let $\mathbb{F}_q$ be a finite field of order q over which an elliptic curve E is defined. The order of a finite field is the number of elements in the field. In a finite field $\mathbb{F}_q$, if q is the prime power, i.e., $q = p^m$, q is the order of field $\mathbb{F}_q$, p is a prime number called the *characteristic* and m is a positive integer. If $m = 1$, then, $\mathbb{F}_q$ is called the *prime field* and if $m \geq 2$, then, $\mathbb{F}_q$ is called the extension field. The number of points in $E(\mathbb{F}_q)$, denoted by $\#E(\mathbb{F}_q)$, is called the *order* of E over $\mathbb{F}_q$.

Theorem 3.2.1. (*Hasse Theorem*) *Let E be an elliptic curve over the finite field $\mathbb{F}_q$, then the order of $E(\mathbb{F}_q)$ satisfies the condition,*

⁴For the full detailed proof of the associativity, please refer to [7]

$$q + 1 - 2\sqrt{q} \leq \#E(\mathbb{F}_q) \leq q + 1 + 2\sqrt{q} \quad (25)$$

This interval, $[q + 1 - 2\sqrt{q}, q + 1 + 2\sqrt{q}]$ is called the *Hasse interval*. This includes the point at infinity. The number of points on the curve roughly increases along with the increase in the number of points in the field.

Group Structure

From Theorem 3.2.1, we can say that if an elliptic curve E is defined over $\mathbb{F}_q$, $E(\mathbb{F}_q)$ is a finite abelian group. It is always cyclic or the product of two cyclic groups. When two finite fields have order q , they are *isomorphic*.

Theorem 3.2.2. (*Group structure of an elliptic curve*) Let elliptic curve E be defined over finite field $\mathbb{F}_q$. Then, $E(\mathbb{F}_q)$ is isomorphic to $\mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2}$ or, $E(\mathbb{F}_q) \simeq \mathbb{Z}_n$ for some integers $n_1, n_2 \geq 1$ with $n_1 \mid n_2$ and $n_1 \mid q - 1$, or for some integer $n \geq 1$.⁵

For example, Let's take a finite field, $\mathbb{F}_{71}$ and define an elliptic curve, $y^2 = x^3 - x$ over it. This elliptic curve defined over field $\mathbb{F}_{71}$ has 72 points, that is, 71 affine points and the point at infinity. More precisely, we can define these set of points as $\{(x, y) \mid x, y \in \mathbb{F}_{71}, y^2 = x^3 - x\} \cup \{\infty\}$ where ∞ is the point at infinity. The group structure would be given by $\mathbb{Z}_2 \oplus \mathbb{Z}_{36}$ over this field.

The number of points on an elliptic curve or its order defined over a finite field can be computed using the *Schoof's Algorithm* [6, 9].

The elliptic curve over a finite field gives limits to the length of the x and y coordinates. In the finite field, the equation of an elliptic curve can be transformed as,

$$y^2 = (x^3 + ax + b) \pmod{p} \quad (26)$$

where, p is a prime number. The same operations of addition and doubling can be performed on the elliptic curve in a finite field. The elliptic curves used in Elliptic Curve Cryptography is usually defined over a finite field.

⁵The definitions, theorems and the Group Law statement in this section (3) is referred from [3, 6, 7]

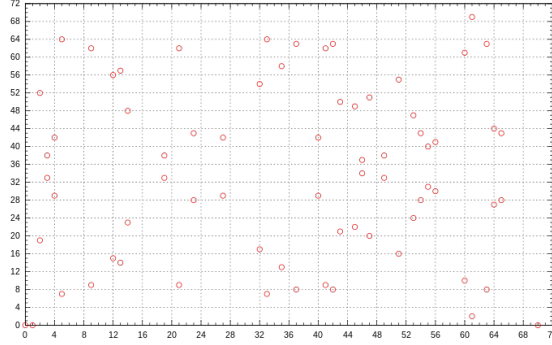


Figure 4: Points of elliptic curve $y^2 = x^3 - x$ over finite field $\mathbb{F}_{71}$

4 Elliptic Curve Cryptography

In 1985, Neal Koblitz [10] and Victor S. Miller [11] independently proposed the use of elliptic curves in public key cryptosystems. This cryptography uses the concepts of elliptic curves and the group law.

4.1 Key Generation

Let E be an elliptic curve that is defined over the finite field $\mathbb{F}_p$. Let P be a point in $E(\mathbb{F}_p)$ which has a prime order, n . From point doubling under the group law and successive doubling, we can say that P generates a cyclic subgroup such that,

$$\langle P \rangle = \{\infty, P, 2P, 3P, 4P, \dots, (n-1)P\} \quad (27)$$

As explained under public key encryption in section 2.2, we know that the sender and receiver each has a private and public key. So, the generator point P , its order n , prime p , and the equation of the elliptic curve E are public domains. The sender or receiver's *private key* is an integer ' k ' which is selected randomly from the interval $[1, n-1]$. And, his or her corresponding *public key* is point Q on the same elliptic curve such that, $Q = kP$. The x and y coordinates of point Q are provided as the public key.

4.2 Encryption scheme

Let's say that Alice wants to send a message, M , to Bob. They decide to use Elliptic Curve Cryptography for the security of their message. Then they

choose to use an elliptic curve E which contains a point P as a generator with order n . Alice chooses a random integer a from the interval $[1, n - 1]$ as her private key and Bob randomly chooses an integer b from the same interval as his private key.

From the key generation just discussed previously (4.1), we know that Alice's public key is $R = aP$ and Bob's public key is $Q = bP$. Alice computes a secret key, using her private key and Bob's public key, $a(Q) = a(bP) = (ab)P$. Bob also computes a secret key using his private key and Alice's public key, $b(R) = b(aP) = (ab)P$.

Hence, $(ab)P$ is the shared secret key between Alice and Bob. Alice encrypts her message using this secret key and sends it to Bob.

$$C_1 = M + (ab)P$$

Bob uses the same shared key to decrypt the message.

$$C_2 = C_1 - (ab)P = M$$

Thus, Alice and Bob have established a secure connection using Elliptic Curve Cryptography.

5 Conclusion and Further Directions

Hence, using the concepts of Groups over finite fields in Algebra, Elliptic Curves, Point addition, Point doubling and the Group Law, today we have a strong cryptography that is secure from many attacks, thus, helping us to share messages confidentially.

When the sizes of various parameters of different cryptographic protocols are compared, it can be noticed that Elliptic Curve Cryptography or ECC contains smaller parameters and shorter key sizes. Having a smaller key size with the same or higher level of security as other protocols make computations in ECC much faster and more efficient with lower power consumption and memory usage.

‘Curve25519’ is one example out of many elliptic curves used in ECC. It has 128 bits security (256 bits key size). This curve has equation $y^2 = x^3 + 486662x^2 + x$ over a prime field defined by prime number $2^{255} - 19$. It’s generator is $x = 9$ which generates a cyclic subgroup of prime order $2^{252} + 27742317777372353535851937790883648493$. The Bitcoin cryptocurrency also uses ECC for it’s security.

Every cryptographic protocol contains an underlying mathematical problem. The strength of the security depends on the difficulty level of this mathematical problem. For ECC, this problem is called the **Elliptic Curve Discrete Logarithmic Problem** or **ECDLP** [3, 6]. This problem tries to find out that given all the public parameters such as the elliptic curve used, the generator P of order n , prime p and the public keys of the users, if it’s possible to compute the user’s private keys and hence, the shared secret key. The difficulty of the problem impacts performance as it dictates parameters such as key and domain sizes. And this in turn dictates the performance of the arithmetic operations.

Many theoretical and generic solutions have been proposed to solve the ECDLP such as the Pollard’s ρ Algorithm, the MOV attack, the Anomalous Attack, Shanks’ Baby Step/ Giant Step, etc. [3, 4, 6, 7] The Pollard’s ρ Algorithm was the known to solve the ECDLP for an elliptic curve over a 109-bit prime field [3]. But that was the largest size it could solve. That is, curves such as ‘Curve25519’ prevented the attack. Thus, none of the attacks would be effective for a carefully chosen elliptic curve and finite field. Many experts believe that other cryptographic algorithms could be broken soon, leaving ECC as the only reasonable alternative.

The current research in this field is trying to find out ways in which ECC could be vulnerable to any attack or ways the ECDLP could be solved. So far, there is a possibility that ECC could be broken by using *Quantum Computing* [12], although it hasn’t been practically proven yet.

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